

Controllable Generation With Text-to-Image Diffusion Models: A Survey

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(Survey Paper)

Abstract—In the rapidly advancing realm of visual generation, diffusion models have revolutionized the landscape, marking a significant shift in capabilities with their impressive text-guided generative functions. However, relying solely on text for conditioning these models does not fully cater to the varied and complex requirements of different applications and scenarios. Acknowledging this shortfall, a variety of studies aim to control pre-trained text-to-image (T2I) models to support novel conditions. In this survey, we undertake a thorough review of the literature on controllable generation with T2I diffusion models, covering both the theoretical foundations and practical advancements in this domain. Our review begins with a brief introduction to the basics of denoising diffusion probabilistic models (DDPMs) and widely used T2I diffusion models. Additionally, we provide a detailed overview of research in this area, categorizing it from the condition perspective into three directions: generation with specific conditions, generation with multiple conditions, and universal controllable generation. For each category, we analyze the underlying control mechanisms and review representative methods based on their core techniques.

Index Terms—Survey, text-to-image diffusion model, controllable generation, AIGC.

I. INTRODUCTION

DIFFUSION models, representing a paradigm shift in the visual generation, have dramatically outperformed traditional frameworks like Generative Adversarial Networks (GANs) [1], [2], [3], [4]. As parameterized Markov chains, diffusion models exhibit a remarkable ability to transform random noise into intricate images, progressing sequentially from noise to high-fidelity visual representations. With the advancement of technology, diffusion models have demonstrated immense potential in image generation and related downstream tasks.

As the quality of imagery generated by these models advances, a critical challenge becomes increasingly apparent: achieving precise control over these generative models to fulfill

complex and diverse human needs. This task goes beyond simply enhancing image resolution or realism; it involves meticulously aligning the generated output with the user's specific and nuanced requirements as well as their creative aspirations. Fueled by the advent of extensive multi-modal text-image datasets [5], [6], [7], [8] and development of guidance mechanism [9], [10], [11], [12], text-to-image (T2I) diffusion models have emerged as a cornerstone in the controllable visual generation landscape [12], [13], [14], [15], [16], [17]. These models are capable of generating realistic, high-quality images that accurately reflect the descriptions provided in natural language.

While text-based conditions have been instrumental in propelling the field of controllable generation forward, they inherently lack the capability to fully satisfy all user requirements. This limitation is particularly evident in scenarios where conditions, such as the depiction of an unseen person or a distinct art style, are not effectively conveyable through text prompts alone. These scenarios pose significant challenges in the T2I generation process, as the nuances and complexities of such visual representations are difficult to encapsulate in text form. Recognizing this gap, a substantial body of research has shifted focus towards integrating novel conditions that extend beyond the confines of textual descriptions into T2I diffusion models. This pivot has been further facilitated by the emergence of powerful and open-sourced T2I diffusion models, as illustrated in Fig. 1(a). These advancements have led to the exploration of diverse conditions, thereby enriching the spectrum of possibilities for conditional generation and addressing the more intricate and nuanced demands of users in various applications.

There are numerous survey articles exploring the AI-generated content (AIGC) domain, including diffusion model theories and architectures [19], efficient diffusion models [20], multi-modal image synthesis and editing [21], visual diffusion model [22], [23], and text-to-3D applications [24]. However, they often provide only a cursory brief of controlling text-to-image diffusion models or predominantly focus on alternative modalities. This lack of in-depth analysis of the integration and impact of novel conditions in T2I models highlights a critical area for future research and exploration.

This survey presents a comprehensive review of controllable generation with text-to-image diffusion models, covering both theoretical foundations and practical advancements. We begin with a concise overview of T2I diffusion models, introducing a brief summary of the theory and widely adopted text-to-image

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For an exhaustive list of the controllable generation literature surveyed, please refer to our curated repository at <https://github.com/PRIV-Creation/Awesome-Controllable-T2I-Diffusion-Models>.

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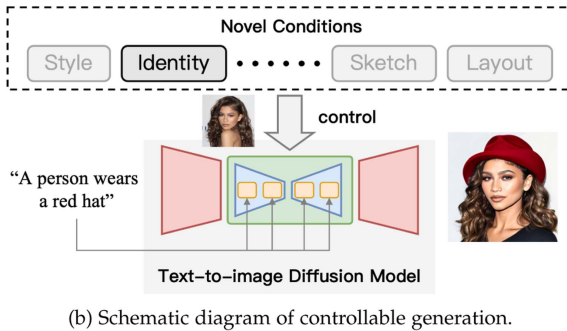
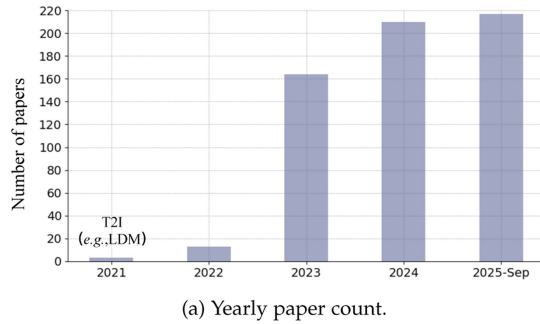


Fig. 1. An overview of conditional generation with T2I diffusion model. (a) We plot the number of papers on controllable generation based on T2I diffusion models, implying that it is increasing rapidly after powerful generators are released. (b) We present a schematic illustration of controllable generation using the T2I diffusion model, where novel conditions beyond text are introduced to steer the outcomes. Example images are sourced from [18].

models. We then provide an in-depth examination of prior studies from a technical perspective, analyzing their theoretical underpinnings and highlighting their distinctive contributions and characteristics. This discussion not only clarifies the foundations of earlier research but also deepens the understanding of the field. Furthermore, we review the diverse applications of these methods, demonstrating their practical value and influence across different contexts and related tasks.

In summary, our contributions are:

- We introduce a well-structured taxonomy of controllable generation methods from the condition perspective, encompassing novel condition introduction, multi-condition integration, and universal controllable generation. This taxonomy provides a clearer lens to reveal their core theoretical principles as well as the inherent challenges of each category.
- We summarize two fundamental paradigms for incorporating novel conditions into T2I diffusion models: conditional score prediction and condition-guided score estimation. Based on these paradigms, we systematically organize and review the corresponding methods, covering a broad spectrum of controllable generation studies. We carefully highlight the key features, distinctive contributions, and comparative advantages of each method.
- We further showcase the diverse applications of conditional generation with T2I diffusion models across a variety of generative tasks, illustrating its emergence as a central and influential component in the AIGC era.

The rest of this paper is organized as follows. Section II provides a brief introduction to the theory and widely used text-to-image diffusion models, overviews the popular controllable generation tasks, and presents a well-structured taxonomy. Later, we summarize existing approaches for controlling the text-to-image diffusion model according to our proposed taxonomy, including novel condition introduction (Section III), multi-condition integration (Section IV), and universal controllable generation (Section V). Finally, Section VI demonstrates the applications of controllable text-to-image generation.

II. PRELIMINARIES

A. Denoising Diffusion Probabilistic Models

Denoising Diffusion Probabilistic Models (DDPMs) represent a novel class of generative models that operate on the principle of reverse diffusion. These models are formulated as parameterized Markov chains that synthesize images by gradually converting noise into structured data through a sequence of steps.

1) *Forward Process*: The diffusion process begins with the data distribution $x_0 \sim q(x_0)$ and adds Gaussian noise incrementally over T timesteps. At each step t , the data x_t is noised by a transition kernel:

$$q(x_{1:T}|x_0) := \prod_{t=1}^T q(x_t|x_{t-1}), \quad (1)$$

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t\mathbf{I}), \quad (2)$$

where β_t are variance hyperparameters of the noise.

2) *Reverse Process*: During the reverse process of a DDPM, the model's objective is to progressively denoise the data, thereby approximating the reverse of the Markov chain. This process begins from the noise vector x_T and transitions towards the original data distribution $q(x_0)$. The generative model parameterizes the reverse transition $p_\theta(x_{t-1}|x_t)$ as a normal distribution:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \quad (3)$$

where deep neural networks, often instantiated by architectures like UNet, parameterize the mean $\mu_\theta(x_t, t)$ and variance $\Sigma_\theta(x_t, t)$. The UNet takes the noisy data x_t and time step t as inputs and outputs the parameters of the normal distribution, thereby predicting the noise ϵ_θ that the model needs to reverse the diffusion process. To synthesize new data instances x_0 , we initiate by sampling a noise vector $x_T \sim p(x_T)$ and then successively sample from the learned transition kernels $x_{t-1} \sim p_\theta(x_{t-1}|x_t)$ until we reach $t = 1$, completing the reverse diffusion process.

Subsequent research has extended the DDPM framework toward more general and theoretically grounded formulations, including the Denoising Diffusion Implicit Model (DDIM) [25], score-based generative modeling [10], Flow Matching [26], and Elucidated Diffusion Models (EDM) [27]. Among these, we focus on the flow-matching technique, as it underpins many state-of-the-art text-to-image systems such as Stable Diffusion 3 [28] and Flux [29].

TABLE I

COLLECTION OF PRIMARY AND USED TEXT-TO-IMAGE DIFFUSION MODELS IN THIS SURVEY. $\dagger$: NUMBER OF UNET AND TEXT ENCODER'S PARAMETERS (DEFAULT REFERS ONLY TO UNET). f : DOWNSAMPLING FACTOR OF AUTOENCODER IN LATENT-SPACE DIFFUSION MODELS. CLIP: OPEN SOURCE IMPLEMENTATION OF CLIP. *: TRAIN FROM SCRATCH. **RESOLUTION**: THE MAXIMUM SUPPORTED IMAGE RESOLUTION GENERATED BY THE MODEL.

Model	Pub.	Param.	Resolution	f	Text Encoder	Arch.	Training Dataset	Open
<i>Pixel Space Diffusion Models</i>								
GLIDE [12]	ICML 2022	5.0B $\dagger$	256 ²	-	plain Transformer* [31]	U-Net	DALL-E [13]	✓
Imagen [15]	NeurIPS 2022	3.0B	1024 ²	-	T5-XXL [32]	U-Net	>LAION-400M [7]	✗
DALL-E 2/3 [33]	arXiv 2022	4.5B	1024 ²	-	CLIP* [34] & Diffusion prior*	U-Net	CLIP [34] & DALL-E [13]	✗
DeepFloyd IF [35]	-	1.5B~6.2B	1024 ²	-	T5-XXL [32]	U-Net	>LAION-A 1B [7]	✓
<i>Latent Space Diffusion Models</i>								
LDM [14]	CVPR 2022	903M	256 ²	8	BERT-tokenizer [36]	U-Net	LAION-400M [7]	✓
SD 1.x [14]	CVPR 2022	860M	512 ²	8	CLIP-L [34]	U-Net	LAION-2B [8]	✓
SD 2.x [14]	CVPR 2022	865M	512 ² /768 ²	8	CLIP-H/14 [34]	U-Net	LAION-5B [8]	✓
SD XL [16]	ICLR 2024	2.6B	1024 ²	8	CLIP-G & CLIP-L [34]	U-Net	internal dataset	✓
SD 3.x [28]	ICML 2024	2B~8.1B	2048 ²	8	CLIP-G&L [34] & T5-XXL [32]	MM-DiT	internal dataset	✓
PixArt- α [37]	ICLR 2024	0.25B	1024 ²	8	T5-XXL [32]	DiT	mixture	✓
Flux.1x [29]	arXiv 2025	12B	2048 ²	8	CLIP-L [34] & T5-XXL [32]	MM-DiT	internal dataset	✓

B. Flow Matching and Rectified Flow

1) *Flow Matching (FM)* [26]: Flow Matching provides a deterministic reformulation of diffusion-based generative modeling by directly learning a continuous vector field that transports probability mass from the data distribution to a simple prior. Let x_t denote the data state at time $t \in [0, 1]$ evolving under

$$\frac{dx_t}{dt} = u_t(x_t), \quad x_0 \sim p_{\text{data}}, \quad x_1 \sim p_1, \quad (4)$$

where $v_t(x)$ represents the velocity field. To enable supervised training, a conditional Gaussian path $p_t(x | x_0)$ is defined with analytical velocity

$$u_t(x_t | x_0) = \frac{\dot{\sigma}_t}{\sigma_t} (x_t - \mu_t) + \dot{\mu}_t, \quad (5)$$

parameterized by time-dependent schedules μ_t and σ_t (which may depend on x_0). A neural network $u_\theta(x_t, t)$ is trained to approximate this target velocity via the *flow matching loss*:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{x_0, \epsilon, t} [\|u_\theta(x_t, t) - u_t(x_t | x_0)\|^2], \quad (6)$$

where $x_t = \mu_t(x_0) + \sigma_t \epsilon$. Here, $\mu_t(x_0)$ denotes a deterministic interpolation function, distinct from the learnable mean $\mu_\theta(x_t, t)$ used in DDPMs. After training, generation proceeds deterministically by integrating the learned ODE from noise to data, removing stochasticity and improving sampling efficiency compared with DDPMs.

2) *Rectified Flow (RF)* [30]: Rectified Flow simplifies the FM formulation by adopting a straight-line interpolation between the prior and the data:

$$x_t = (1 - t)x_1 + tx_0, \quad u_t(x_t | x_0) = x_0 - x_1, \quad (7)$$

with $x_0 \sim p_{\text{data}}$, $x_1 \sim p_1$, and $t \sim \mathcal{U}(0, 1)$. The training objective remains a mean-squared error on the predicted velocity:

$$\mathcal{L}_{\text{RF}} = \mathbb{E}_{x_0, x_1, t} [\|u_\theta(x_t, t) - (x_0 - x_1)\|^2]. \quad (8)$$

This rectification enforces a monotonic and low-curvature probability flow, resulting in smoother vector fields and faster ODE integration.

C. Text-to-Image Diffusion Models

Text-to-image (T2I) diffusion models synthesize images from textual descriptions by learning a conditional denoising process. A central design question in such models is how textual information is injected into the denoising network. We highlight here the representative architectural paradigms that define modern T2I systems regarding task formulation and architecture-level analysis (Table I).

1) *Cross-Attention-Based U-Net Architectures*: Most early and widely adopted T2I models employ a U-Net backbone augmented with cross-attention layers to model. In this mechanism, latent image features act as queries, while text embeddings provide keys and values, enabling fine-grained alignment between linguistic concepts and visual features. GLIDE [12] first demonstrated that replacing class-conditioning in diffusion models with free-form text, combined with classifier-free guidance (CFG) [11], yields strong improvements in photorealism and text alignment. Imagen [15] further showed that scaling the text encoder (a frozen large language model, e.g., T5 [32]) improves generation quality more than enlarging the diffusion model itself, and confirmed cross-attention as the most effective conditioning approach. Latent Diffusion Models (LDM) [14] introduced a major efficiency breakthrough by performing diffusion in a compressed latent space, enabling high-resolution synthesis on limited compute. Stable Diffusion (SD) [14] builds on the LDM formulation and its v1.x, v2.x, and SDXL variants adopt the U-Net with cross-attention design that has become the de facto standard in open-source T2I generation.

2) *Transformer-Based Diffusion Models: DiT and MMDiT*: Recent T2I systems, such as Stable Diffusion 3 [28] and FLUX [29], increasingly replace U-Nets with transformer-based diffusion backbones due to their improved scalability and expressiveness. Diffusion Transformers (DiT) [38] treat image latents as patch tokens and perform denoising entirely through self-attention, showing superior scaling behavior compared to convolutional models. Beyond traditional cross-attention, Multimodal Diffusion Transformers (MMDiT) introduce *joint attention* mechanisms in which text and image tokens interact

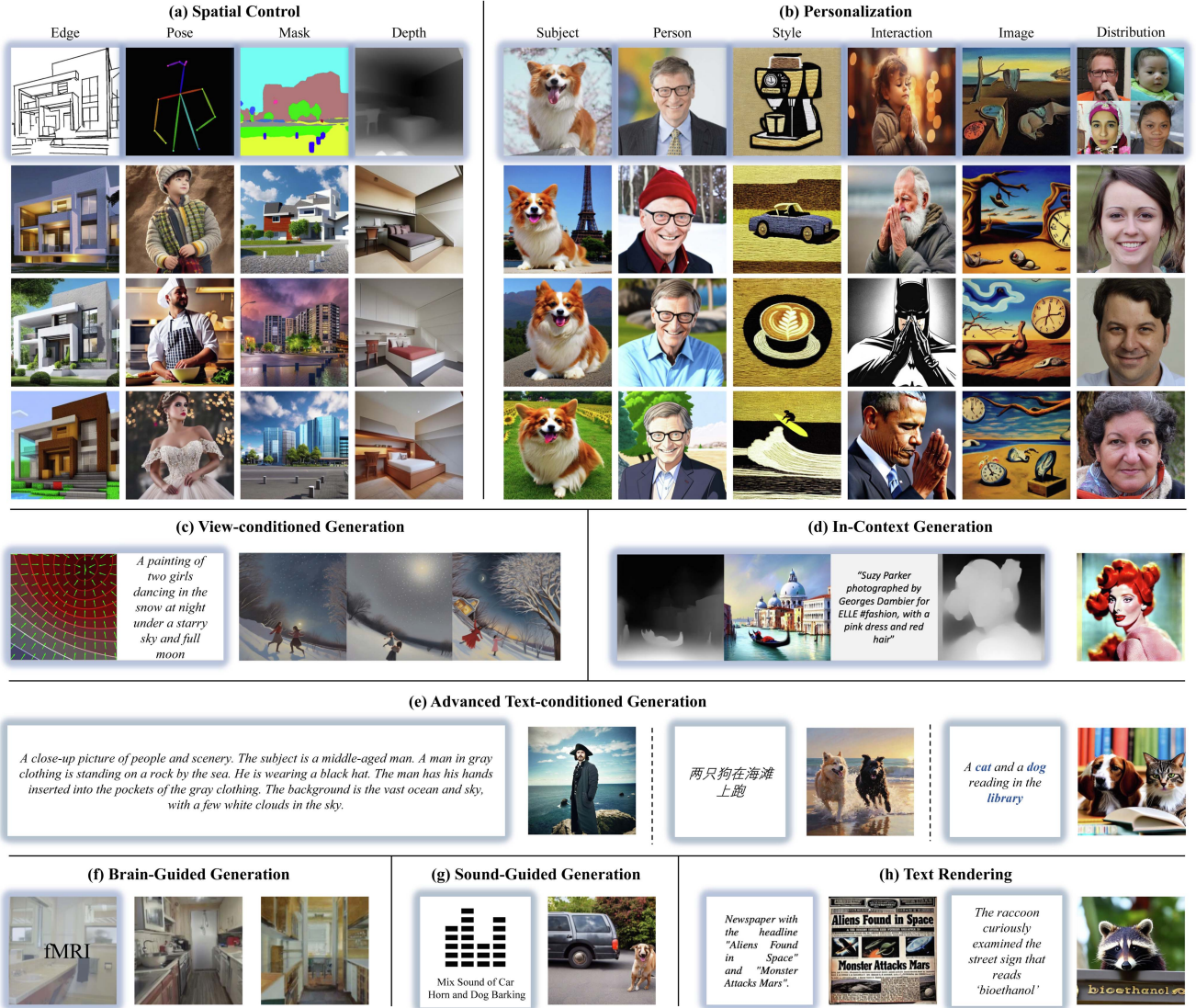


Fig. 2. Illustration of controllable text-to-image generation with specific conditions. The condition is marked in blue background. Examples are sourced from [18], [33], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50].

bidirectionally within the same transformer blocks. This unified multimodal modeling enables richer text-image dependency structures than unidirectional cross-attention.

D. Controllable Generation Tasks

Controllable generation extends the capabilities of text-to-image diffusion models by introducing diverse conditions that guide the synthesis process beyond plain textual prompts. These conditions enable more precise alignment with user intentions. In this part, we introduce the major categories of controllable generation tasks and illustrate them in Fig. 2:

1) *Spatial Control*: Since text alone struggles to represent structural information such as position and dense labels, spatial signals have become crucial conditions for text-to-image diffusion. Typical spatial inputs include layout, human pose, depth, and segmentation masks.

2) *Image Personalization*: The personalization task aims to capture and utilize concepts from exemplar images that cannot be easily described by text, integrating them as generative conditions for controllable synthesis.

3) *View-conditioned Generation*: View-conditioned generation aims to synthesize images from specific viewpoints or across multiple views, ensuring geometric consistency and structural coherence. By leveraging conditions such as camera parameters, depth maps, or multi-view correspondences, these methods extend controllable diffusion to scenarios like novel view synthesis, 3D-aware image generation, and panoramic rendering.

4) *Advanced Text-Conditioned Generation*: Although text is the fundamental condition in text-to-image diffusion models, several challenges remain. For example, text-guided synthesis often suffers from misalignment, particularly when dealing with complex prompts involving multiple entities or rich contextual descriptions. Moreover, the dominance of English datasets in

TABLE II
SUMMARY OF EVALUATION DIMENSIONS AND COMMONLY USED METRICS FOR
CONTROLLABLE T2I MODELS

Dimension	Metrics
Image Quality	FID [52]; IS [53]; Aesthetic Score [54]; Human Preference (HPS [55], PickScore [56])
Conditional Alignment	Textual Similarity (CLIP [34], [57]); Visual Similarity (CLIPScore [34], DINO [58]); Spatial Consistency (mIoU, MSE); Content Accuracy [58], [59]

training leads to limited multilingual capabilities. To address these issues, recent works propose novel strategies for improving alignment and expanding the linguistic scope of text-conditioned generation.

5) *In-Context Generation*: In-context generation focuses on understanding and executing specific tasks on query images by leveraging task-specific example pairs of images and text guidance. This setting enables models to adapt to new tasks with minimal supervision, expanding the flexibility of controllable diffusion.

6) *Brain- & Sound-Guided Generation*: Brain-guided generation seeks to control image synthesis directly from neural activity, such as electroencephalogram (EEG) recordings or functional magnetic resonance imaging (fMRI), bypassing the intermediate step of translating thoughts into text. Similarly, sound-guided generation explores how auditory signals can be leveraged as direct conditions for visual synthesis, enabling cross-modal creativity.

7) *Text Rendering*: Rendering coherent and legible text within generated images is a critical task, given its wide applications in posters, book covers, advertisements, and memes. Effective text rendering not only enhances practicality but also pushes the boundary of fine-grained controllability in diffusion models.

E. Evaluation of Controllable T2I Generation

Since the types of conditions used in controllable T2I methods are highly diverse and vary across approaches, we focus here on the overarching evaluation dimensions rather than specific task settings. In what follows, we outline the main perspectives from which controllable T2I models are commonly assessed and summarize the widely adopted metrics associated with each dimension (Table II).

1) *Image Quality*: Image quality is commonly measured by distribution-based metrics such as Inception Score (IS) [52] and Fréchet Inception Distance (FID) [51], computed on deep features from a pretrained classifier (e.g., Inception V3 [60]). In addition, aesthetic predictors [53] and learned human-preference models (HPS [54], PickScore [55]) estimate visual appeal and perceived quality directly from generated images.

2) *Conditional Alignment*: Conditional alignment measures how faithfully the generated images follow the specified conditions (textual or otherwise). Text-image alignment is typically quantified with CLIP-based similarity scores [34], [56], which leverage a large-scale contrastively trained vision-language model to compare prompts and outputs. For other types of

conditions, alignment is computed as the discrepancy between the input condition and its prediction from the generated image, for example by using CLIP or DINOv2 [57] features for image-conditioned tasks, or detection metrics such as mAP for layout-conditioned generation.

F. Taxonomy

Conditional generation with text-to-image diffusion models is inherently complex and can be broadly organized into three sub-tasks from the perspective of conditioning signals. The first and most studied direction augments pretrained diffusion models with novel conditions. We group these methods by their theoretical foundations, namely conditional score prediction and condition-guided score estimation. The main challenge is to flexibly inject new condition types alongside text prompts without sacrificing image quality. The second direction targets multi-condition control, such as combining a character's identity with a specific pose. Methods are categorized by their technical strategies, including joint training, continual learning, weight fusion, attention-based integration, and guidance composition. Here, the core difficulty is to fuse multiple signals so that all conditions are faithfully expressed. The third direction pursues condition-agnostic generation, aiming to build unified frameworks that can robustly leverage diverse condition types across a wide range of inputs.

III. CONTROL TEXT-TO-IMAGE DIFFUSION MODELS WITH NOVEL CONDITIONS

Following [61], we can set the approximate denoising transition mean $\mu_\theta(x_t, t)$ in (3) as:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}x_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t}}s_\theta(x_t, t) \quad (9)$$

where $s_\theta(x_t, t)$ is a neural network that learns to predict the score function $\nabla_{x_t} \log p_t(x)$. Hence, we have:

$$\nabla_{x_t} \log p_t(x) = -\frac{1}{\sqrt{1 - \bar{\alpha}_t}}\epsilon \quad (10)$$

where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ is the Gaussian noise used in forward process, $\alpha_t := 1 - \beta_t$, and $\bar{\alpha}_t := \prod_{s=0}^t \alpha_s$. Then, (9) can be written as:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \hat{\epsilon}(x_t, t) \right) \quad (11)$$

where $\hat{\epsilon}(x_t, t)$ predicts ϵ .

In conditional generation (c denotes condition), the score function is extended with a posterior probability term $\nabla_{x_t} \log p_t(c|x_t)$ and becomes $\nabla_{x_t} \log(p_t(x_t)p_t^w(x_t|c))$ (w represents a hyper-parameter to control condition intensity), following [9], [11]. To employ a neural network for conditional generation, classifier-free guidance (CFG) [11] transforms it to:

$$\begin{aligned} & \nabla_{x_t} \log(p_t(x_t)p_t^w(c|x_t)) \\ &= \nabla_{x_t} \log p_t(x_t) + w \nabla_{x_t} \log p_t(c|x_t) \\ &= \nabla_{x_t} \log p_t(x_t) + w \nabla_{x_t} \log \frac{p_t(x_t|c)}{p_t(x_t)} \end{aligned}$$

$$= (1 - w) \nabla_{x_t} \log p_t(x_t) + w \nabla_{x_t} \log p_t(x_t|c) \quad (12)$$

where $\nabla_{x_t} \log p_t(x_t)$ and $\nabla_{x_t} \log p_t(x_t|c)$ can be predicted by training a model $\epsilon_\theta(x_t, \cdot, t)$, which predict the former via $\epsilon_\theta(x_t, \phi, t)$ and the latter via $\epsilon_\theta(x_t, c, t)$.

Existing T2I diffusion models train $\epsilon_\theta(x_t, \cdot, t)$ by randomly dropping the text prompt, and the denoising process with CFG is as follows:

$$\hat{\epsilon}(x_t, c_{text}, t) = (1 - w) \epsilon_\theta(x_t, \phi, t) + w \epsilon_\theta(x_t, c_{text}, t) \quad (13)$$

and $\hat{\epsilon}(x_t, c_{text}, t)$ is used in (9) for conditional synthesis.

Hence, the key to controlling text-to-image models with novel conditions c_{novel} is to model score $\nabla_{x_t} \log p_t(x_t|c_{text}, c_{novel})$. Following [9], [62], there are two types of mechanisms, *i.e.*, conditional score prediction and conditioned-guided score estimation, which we illustrate below.

- 1) *Conditional Score Prediction (Section III-A)*: While T2I diffusion models leverage $\epsilon_\theta(x_t, c_{text}, t)$ to predict $\nabla_{x_t} \log p_t(x_t|c_{text})$, a fundamental and powerful way for steering diffusion models is through conditional score prediction in the sampling process, where these methods introduce c_{novel} into $\epsilon_\theta(x_t, c_{text}, t)$, constructing a $\tilde{\epsilon}(x_t, c_{text}, c_{novel}, t)$ to straightforwardly predict $\nabla_{x_t} \log p_t(x_t|c_{text}, c_{novel})$. The CFG-based denoising update then reads:

$$\hat{\epsilon}(x_t, c_{text}, c_{novel}, t) = (1 - w) \tilde{\epsilon}(x_t, \phi, t) + w \tilde{\epsilon}(x_t, c_{text}, c_{novel}, t) \quad (14)$$

We here illustrate several mainstream ways to attain $\tilde{\epsilon}(x_t, c_{text}, c_{novel}, t)$.

- 2) *Condition-guided Score Estimation (Section III-B)*: Unlike conditional score prediction straightforwardly predicting $\nabla_{x_t} \log p_t(x_t|c_{text}, c_{novel})$, condition-guided estimation approaches estimate $\log p_t(c_{novel}|x_t)$ with a likelihood/critic and obtain $\nabla_{x_t} \log p_t(c_{novel}|x_t)$ by back-propagation, which is then injected into the sampler. And the denoising process now reads:

$$\hat{\epsilon}(x_t, c_{text}, c_{novel}, t) = \hat{\epsilon}(x_t, c_{text}, t) + \gamma \nabla_{x_t} \log p_t(c_{novel}|x_t) \quad (15)$$

where γ is a hyper-parameter to adjust the conditional score and $\hat{\epsilon}(x_t, c_{text}, t)$ is the original score prediction of text-conditioned diffusion models with CFG.

A. Conditional Score Prediction

Conditional score prediction approaches focus on empowering pre-trained denoising models supporting novel conditions to straightforwardly predict denoised latents. According to mechanisms, these methods can be categorized into tuning-based (Section III-A1), adapter-based (Section III-A2), and training-free (Section III-A3) manners.

1) *Tuning-Based Conditional Score Prediction*: Tuning-based methods typically focus on adapting to a specific condition, often in scenarios with limited data, such as single or few-shot examples. These methods achieve conditional prediction by transforming either the text condition c_{text} or the model

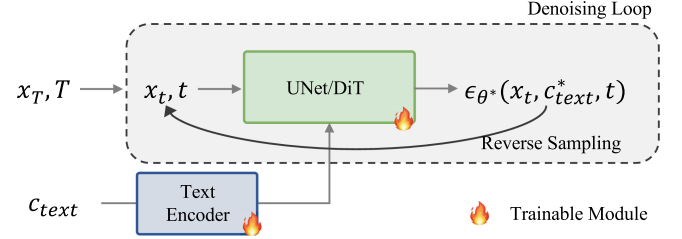


Fig. 3. Illustration of tuning-based conditional score prediction.

parameters θ into a form specific to the given condition, as shown in Fig. 3. This can be represented as:

$$\tilde{\epsilon}(x_t, c_{text}, c_{novel}, t) = \epsilon_{\theta^*}(x_t, c_{text}^*, t) \quad (16)$$

where condition information is memorized in c_{text} and θ .

a) *Basic tuning-based methods*: A straightforward yet effective strategy for learning controlling concepts from given samples is to fine-tune the diffusion models, thereby adapting text-to-image models to reflect the target conditions. The updated parameters are specialized to capture the desired conditions [63], [64], [65], [66], [67], [68], [69], [70].

As the fundamental input of text-to-image diffusion models, text plays a central role in adapting these models to user-specific requirements. Textual Inversion (TI) [64] introduces an innovative idea by embedding user-provided concepts into new “words” in the text embedding space. This expands the tokenizer’s dictionary and optimizes additional tokens through a denoising process applied on provided images. DreamBooth [65] adopts a similar approach but encodes concepts with low-frequency words (*i.e.*, *sks*), and additionally updates the UNet parameters with a class-specific prior preservation loss to enhance output diversity. Together, the simplicity and adaptability of TI and DreamBooth have established them as foundational frameworks for many subsequent tuning-based methods. Building on these foundations, Custom Diffusion [71] examines weight deviations during fine-tuning and identifies cross-attention layer parameters—particularly the key and value projections (W^k and W^v)—as pivotal. It thus narrows updates to these projections and augments the process with extra text tokens and a regularization loss.

To further improve textual inversion, recent works explore layer-specific distinctions in UNet [66], [67], [72], refine embedding initialization strategies [73], and sampling distribution [74]. These methods apply distinct text embeddings across layers to capture finer variations. In contrast, CatVersion [75] moves away from tuning text embeddings or UNet parameters, and instead learns concatenated embeddings within the dense feature space of the text encoder. This design proves effective for capturing subtle differences between a personalized concept and its base class, thereby helping preserve prior knowledge.

b) *Parameter-efficient fine-tuning (PEFT)*: Beyond full fine-tuning, parameter-efficient methods (PEFT) [76], [77], [78], [79] have become increasingly important in personalization [80]. Among them, Low-rank Adaptation (LoRA) [77] has been widely adopted in various personalization pipelines [65], [81],

[82], [83], [84], [85]. Xiang et al. propose ANOVA [86], which employs adapters [76] and demonstrates that placing them after the cross-attention block notably boosts performance. COM-CAT [80] develops an efficient ViT [87] compression method based on model factorization. Similarly, DiffuseKronA [85] introduces a Kronecker product-based adaptation module that surpasses LoRA-DreamBooth in parameter efficiency and stability, offering consistent high-quality generation with greater interpretability. To support research and evaluation in this area, LyCORIS [88] provides a comprehensive open-source library¹ covering numerous PEFT methods (e.g., LoRA, LoHa, Dy-LoRA [78]) and a framework for their systematic assessment, thus promoting the progress of diffusion model personalization.

c) Condition disentanglement: A further challenge in introducing novel conditions lies in disentangling the desired concept from confounding inputs. Many studies [81], [84], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101], [102] observe that irrelevant information—such as background context or co-occurring objects—tends to be entangled with the target concept. To address this, several works [92], [101], [103] employ explicit masks to isolate object regions. In this direction, DisenBooth [81] and DETEX [93] reduce the impact of backgrounds, with DETEX further decoupling pose from subject appearance. PACGen [94] instead applies aggressive data augmentation, altering the size and position of concepts to help separate spatial cues from the core identity.

Other approaches disentangle conditions by adjusting fine-tuning strategies. DreamVideo [104] separates subject learning and motion learning, while B-LoRA [105] jointly optimizes LoRA weights of two blocks to implicitly decouple style and content. ReVersion [106] introduces a relation-steering contrastive learning scheme to capture object relationships more effectively.

Training techniques can also be exploited to reduce condition entanglement [41], [107], [108]. Selective Information Description [107] leverages a VLM to produce refined text descriptions, ensuring the model emphasizes target objects over contextual biases. Similarly, U-VAP [108] adopts a decoupled self-augmentation strategy, where an LLM generates paired target and non-target prompts, facilitating dual concept learning to decouple conditions.

d) Prior preservation: Another important challenge in fine-tuning diffusion models is the preservation of prior knowledge. Without careful design, models risk overfitting to narrow concepts, sacrificing generality and controllability. To address this, prior preservation techniques have been widely explored [65], [71], [109], [110], [111], [112], [113], [114]. Perfusion [109] mitigates overfitting by locking cross-attention keys to prior categories and applying a gated rank-1 concept update. SVD-iff [111] regulates singular values in weight matrices to reduce risks such as language drift, while OFT [110] introduces orthogonal fine-tuning to preserve semantic capacity by maintaining hyperspherical energy. Together, these methods help balance fidelity to new data with the retention of broad generative ability.

e) Training techniques: Beyond the strategies above, alternative training techniques have been explored to improve efficiency, reduce computational overhead, and enhance performance [75], [115], [116], [117], [118], [119], [120]. For example, DVAR [115] proposes a variance-based early stopping criterion to replace unreliable convergence metrics, thereby accelerating training. Gradient-Free Textual Inversion [116] divides optimization into dimensionality reduction and non-convex gradient-free search, achieving faster convergence with minimal performance loss. MATTE [117] investigates the roles of timesteps and UNet layers for different concept categories, aiming for broader adaptability.

Another notable direction is the introduction of auxiliary losses to boost generation performance [121], [122], [123], [124]. TokenCompose [121] improves multi-object composition and photorealism through token-wise consistency losses between images and segmentation maps. Similarly, CoMat [122] addresses text-image misalignment via a concept activation loss and enhances visual quality with adversarial loss.

Lastly, data-centric strategies have been proposed to enhance the training process. COTI [118] improves Textual Inversion through active and controllable data selection, while He et al. [119] generate text- and image-level regularization datasets to better preserve model generalization.

f) Inference techniques: In addition to training methods, several approaches enhance controllability during inference by modulating cross-attention to better align outputs with target conditions [125], [126], [127], [128]. For example, MagicTailor [125] combats semantic pollution and imbalance with dynamic masked degradation and dual-stream balancing. OMG [129] provides an occlusion-friendly framework that adopts a two-stage sampling process—first generating layout and visual comprehension for occlusion handling, then applying noise blending to integrate concepts—leading to superior identity preservation and visual harmony.

Additionally, DreamBlend [130] addresses the trade-off between prompt fidelity, subject fidelity, and diversity by leveraging multiple checkpoints and combining their strengths through cross-attention guidance. Prompt-aligned personalization [131] improves complex text alignment via score distillation sampling while supporting multi-/single-shot scenarios, subject composition, and reference-guided generation.

2) Adapter-Based Conditional Score Prediction: To eliminate test-time tuning cost, a class of methods introduces an additional encoder E that maps novel conditions to feature embeddings and feeds them into the noise predictor. The conditional score prediction then reads

$$\tilde{\epsilon}(x_t, c_{text}, c_{novel}, t) = \epsilon_{\theta^*}(x_t, c_{text}, E(c_{novel}), t), \quad (17)$$

where θ^* and E are learned *offline* and kept fixed at inference. Fig. 4 gives a schematic overview.

Adapter-based variants primarily differ by the condition families they target (e.g., geometry, style, layout, controls), which dictate how the introduced adapter $E(\cdot)$ interfaces with the backbone (e.g., concatenation to token embeddings, cross-attention keys/values, or feature-wise affine modulation). In what follows,

¹<https://github.com/KohakuBlueleaf/LyCORIS>

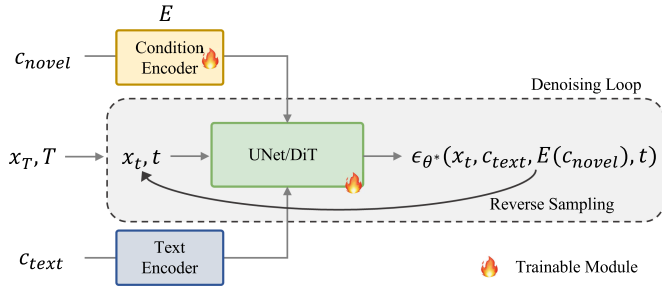


Fig. 4. Illustration of adapter-based conditional score prediction.

we group methods by task type and provide a systematic review; task definitions are summarized in Section II-D.

a) Spatial control: ControlNet [43] stands out among generalized spatial controllers, earning recognition as a seminal work and winning the Marr Award in 2023. Distinct from approaches that simply fine-tune the base diffusion model parameters [71], [132], ControlNet introduces an auxiliary encoder branch mirroring the U-Net and couples it to the original layers via zero convolutions to mitigate overfitting and catastrophic forgetting. Owing to its simplicity and adaptability, ControlNet has proved effective and has become a widely adopted baseline in subsequent studies [133], [134], [135], [136], [137], [138], [139], [140]. Similarly, T2I-Adapter [141] aligns internal knowledge in text-to-image diffusion models with external spatial control signals.

While ControlNet [43] requires training a separate model for each control type, subsequent work seeks general controllers that handle diverse spatial signals [133], [135], [136], [142], [143], [144], [145]. Qin et al. [135] propose UniControl, a task-aware HyperNetwork that modulates the diffusion backbone across condition types: conditions are encoded via a mixture-of-experts (MoE) adapter, while task instructions are embedded by the HyperNet and injected through zero-conv gating to precisely regulate how condition features enter the model. Meta ControlNet [136] adapts the pretrained ControlNet to another condition domain by meta learning, apparently reducing the learning steps. In parallel, Ctrl-Adapter [144] and X-Adapter [146] transfer diverse controls to arbitrary diffusion backbones by adapting pretrained ControlNets.

For layout-conditioned generation, many methods arrange regions and bind them to textual concepts [134], [147], [148], [149], [150], [151], [152], [153], [154], [155], [156]. GLIGEN [147] grounds language with structured inputs and injects the grounding via gated trainable layers, enabling controllable placement. SpaText [151] builds a spatio-textual representation by stacking CLIP-derived object embeddings in masks at their target positions to enforce layout. Related efforts focus on the face domain, synthesizing faces under face-parsing constraints [157], [158].

From the objective perspective, several approaches jointly denoise dense spatial conditions to improve alignment. JointNet [159] augments a pretrained T2I backbone with a dense-modality branch (e.g., depth) that is tightly coupled to the RGB branch, enabling rich cross-modality interactions. Liu

et al. [160] propose a Latent Structural Diffusion Model that co-denoises depth and surface normals alongside RGB synthesis. Complementarily, adding auxiliary spatial-consistency losses further enhances control [161], [162], [163]. Specifically, ControlNet++ [161] enforces pixel-level cycle consistency between outputs and controls, while Li et al. [163] introduce a segmentation-based discriminator to provide explicit spatial feedback.

While many aim for strict adherence to provided controls, other methods support coarse or incomplete spatial inputs [142], [164], [165]. Specifically, LooseControl [164] extracts proxy depth from images to define 3D box controls and fine-tunes ControlNet [43] via LoRA [77], enabling the creation of complex environments (e.g., rooms, street scenes) by specifying only scene boundaries and key object locations.

b) Image personalization: Adapter-based image personalization methods employ encoders to embed concepts (e.g., subject, human identity, style), offering a significant speed advantage over tuning-based approaches when extracting concepts from images.

Some methods adopt a domain-agnostic strategy, training encoders on open-world images to extract generalized subject-level conditions [39], [166], [167], [168], [169], [170], [171], [172], [173], [174], [175], [176]. These methods typically leverage large pretrained encoders such as CLIP [34] and BLIP-2 [177], fine-tuning only lightweight projection layers [39], [166], [168]. ELITE [39], for example, integrates a global mapping network and a local mapping network based on CLIP [34]. The global network transforms hierarchical image features into multiple text embeddings, while the local network infuses patch features into cross-attention layers for detailed reconstruction. BLIP-Diffusion [178] advances customization by pre-training a BLIP-2 [177] encoder for text-aligned image representation and developing a task for learning subject representations, enabling the generation of novel subject renditions. Following on E4T [179], Arar et al. [168] introduce an encoder for acquiring text embeddings and propose a hypernetwork to predict LoRA-style attention weight offsets in UNet. SuTI [167] takes a unique approach inspired by apprenticeship learning [180], training a vast array of expert models on millions of internet image clusters. The apprentice model is then taught to imitate these experts' behaviors. CAFE [181] build a customization assistant based on pre-trained large language model and diffusion model.

Some works design domain-aware encoders tailored to targeted domains [179], [182], [183]. In person-driven settings, facial images are encoded to provide identity conditions [18], [132], [184], [185], [186], [187], [188], [189], [190], [191], [192], [193], [194], [195], [196], [197]. Face0 [186] and DreamIdentity [187] employ pretrained face recognition encoders [198]; Face0 uses Inception-ResNet-V1 [199], while DreamIdentity introduces a ViT-style M²ID encoder [87]. Beyond multimodal and face-recognition encoders [198], $\mathcal{W}^+$ Adapter [200] and PreciseControl [201] leverage GAN inversion [202], [203] encoders as an alternative identity pathway.

Furthermore, some person-driven methods study the mechanism of combining textual embeddings with identity embeddings [185], [188], [193]. To balance identity preservation and

editability, FastComposer [185] and PhotoMaker [188] fuse text prompts with visual features from reference images; specifically, FastComposer mixes human-related text tokens (e.g., “man”, “woman”) with visual features via a multilayer perceptron, and PhotoMaker applies two MLP layers to fuse image embeddings with the corresponding human-related embedding, then updates the latter with the fused representation. Beyond an identity encoder, some works additionally employ a spatial condition encoder to improve generation quality [204]. This task also benefits from face segmentation masks and skeletal cues obtained from off-the-shelf models or annotations [185], [188], [189], [205], [206]. For example, Stellar [205] uses face masks to remove background during preprocessing, sharpening focus on identity, while other methods leverage face masks to construct [185], [188], [189] or adjust [100] loss functions.

Notably, task-specific encoders can be effective for image personalization. For style-driven generation, several studies use VGG features to better capture low-level style [40], [207]. Prompt-free Diffusion [208] introduces SeeCoder, composed of a backbone encoder, a decoder, and a query transformer, enabling reference images to serve as conditions in lieu of text prompts.

c) View-conditioned generation: Adapter-based view-conditioned generation aims to leverage explicit geometric or panoramic constraints within generative models to ensure spatial consistency across different viewpoints [49], [209], [210], [211], [212], [213], [214], [215]. PreciseCam [49] converts four simple extrinsic and intrinsic camera parameters to a PF-US map and then controls generation by a trained ControlNet. StitchDiffusion [213] extends adapter-based controllable generation to the domain of 360-degree panoramas by fine-tuning a T2I diffusion model with LoRA and introducing a stitching-aware denoising strategy. This approach ensures seamless global geometry and strong generalization for panoramic scene synthesis.

d) Advanced text-conditioned generation: To better extract faithful textual semantics, some works leverage LLM to replace the CLIP text encoder [45], [50], [216]. To improve the textual alignment of a long paragraph (up to 512 words), Wu et al. [45] introduce an informative-enriched diffusion model for paragraph-to-image generation task, termed ParaDiffusion, which employ a large language model (e.g., Llama V2 [217]) to encode long-form text, followed by fine-tuning with LoRA [77] to align text-image feature spaces in generation.

Other methods design an extra pipeline or textual encoders to improve textual controllability [216], [218]. Tailored Visions [216] introduces a prompt rewriting system, leveraging historical user interactions to rewrite user prompts to enhance the expressiveness and alignment of user prompts with their intended visual outputs.

The text encoder is also studied to extend to multilingual version. GlueGen [47] aligns multilingual language models (e.g., XLM-Roberta [219]) with existing text-to-image models, allowing for the generation of high-quality images from captions beyond English. PEA-Diffusion [220] is a proposed simple plug-and-play language transfer method based on knowledge distillation, where a lightweight MLP-like parameter-efficient

adapter with only 6 M parameters is trained under teacher knowledge distillation along with a small parallel data corpus.

e) In-context generation: Wang et al. [44] introduced Prompt Diffusion, a novel approach that is jointly trained over multiple tasks using in-context prompts. This method has shown impressive results in high-quality in-context generation for trained tasks and effectively generalizes to new, unseen vision tasks with relevant prompts. Building upon this, Chen et al. [221] further enhance Prompt Diffusion by incorporating a vision encoder-modulated text encoder. This innovation addresses several challenges, including costly pre-training, restrictive problem formulations, limited visual comprehension, and insufficient generalizability to out-of-distribution tasks. Moreover, Najdenkoska et al. [222] propose a novel framework that separates the encoding of the visual context and preserves the structure of the query images. This results in the ability to learn from the visual context and text prompts, but also from either one of them.

f) Brain-Guided Generation: The brain-guided generation tasks focus on controlling image creation directly from brain activities, such as electroencephalogram (EEG) recordings and functional magnetic resonance imaging (fMRI), bypassing the need to translate thoughts into text. More recently, advancements have been made with the adoption of visual diffusion models, offering enhanced capabilities in accurately translating complex brain activities into coherent visual representations [46], [223], [224], [225], [226], [227], [228].

Chen et al. [223] present a Sparse Masked Brain Modeling with Doubled-Conditioned Latent Diffusion Model (MinD-Vis) for human vision decoding. They first learn an effective self-supervised representation of fMRI data using mask modeling and then augment latent diffusion model with double-conditioning. MindDiffuser [46] is also a two-stage image reconstruction model. In the first stage, the VQ-VAE latent representations and the CLIP text embeddings decoded from fMRI are put into the image-to-image process of Stable Diffusion, which yields a preliminary image that contains semantic and structural information. Then, it utilizes the low-level CLIP visual features decoded from fMRI as supervisory information, and continually adjust the two features in the first stage through backpropagation to align the structural information.

While the above methods reconstruct visual results from fMRI, some approaches choose electroencephalogram (EEG) [227], [228], which is a non-invasive and low-cost method of recording electrical activity in the brain. DreamDiffusion [227] leverages pre-trained text-to-image models and employs temporal masked signal modeling to pre-train the EEG encoder for effective and robust EEG representations. Additionally, the method further leverages a CLIP image encoder to provide extra supervision to better align EEG, text, and image embeddings with limited EEG-image pairs.

g) Sound-guided generation: For sound-guided generation, some works develop an additional audio encoder, utilized to embed input audio into text embedding space or latent feature space to control generation [47], [229], [230]. GlueGen [47] aligns multi-modal encoders such as AudioCLIP with the Stable Diffusion model, enabling sound-to-image generation. Yang

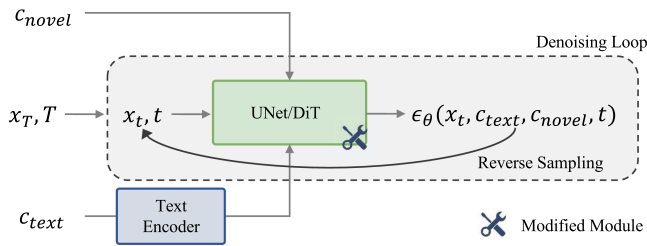


Fig. 5. Illustration of training-free conditional score prediction.

et al. [230] propose a unified framework “Align, Adapt, and Inject” (AAI) for sound-guided image generation, editing, and stylization. In particular, this method adapts input sound into a sound token, like an ordinary word, which can plug and play with existing powerful diffusion-based Text-to-Image models.

h) Text rendering: Drawing inspiration from the analysis in unCLIP [33], which highlights the inadequacy of raw CLIP text embeddings in accurately modeling the spelling information in prompts, subsequent efforts such as eDiff-I [17] and Imagen [15] have sought to harness the capabilities of large language models like T5 [32], trained on text-only corpora, as text encoders in image generation. Additionally, DeepFloyd IF, following the design principles of Imagen [15], has demonstrated impressive proficiency in rendering legible text on images, showcasing a significant advancement in this challenging domain. Meanwhile, some approaches are designed to improve text rendering capability for existing text-to-image diffusion models [48], [231], [232], [233], [234], [235], [236]. GlyphControl [237] leverages additional glyph conditional information to enhance the performance of the off-the-shelf Stable-Diffusion model in generating accurate visual text. TextDiffuser et al. [232] first generates the layout of keywords extracted from text prompts and then generates images conditioned on the text prompt and the generated layout. The authors also contribute a large-scale text images dataset with OCR annotations, MARIO-10 M, containing 10 million image-text pairs with text recognition, detection, and character-level segmentation annotations. Zhang et al. [48] proposed Diff-Text, a training-free scene text generation framework for any language. Diff-text leverages rendered sketch images as priors to render text by ControlNet [43] and proposes a localized attention constraint to address the unreasonable position problem of scene text.

3) Training-Free Conditional Score Prediction: While the above techniques require training, a complementary line of work operates in a training-free fashion (see Fig. 5). These methods analyze and exploit the model’s intrinsic properties so that a pretrained model can directly support novel conditions.

a) Attention-map adjustment: Since attention maps are inherently interpretable, many controllable generation methods build upon attention map adjustment. Attention maps explicitly capture the relationships between tokens, particularly establishing correspondences between image features and textual tokens. This property enables direct control over the placement of textual concepts in generated images, including object positioning (i.e., layout-to-image) [17], [238], [238], [239], [240], [241], [242],

[243], [244], [245], [246], fine-grained control of objects and their attributes (i.e., attribute binding) [247], and improving textual concept representation [248].

For instance, since attention mechanisms explicitly model the relationships between text and image tokens, modulating the attention map becomes a pivotal training-free technique for controlling structure in score prediction [17], [238], [240], [241]. eDiff-I [17] presents a technique named “*paint-with-words*” (also known as pww), rectifying the cross-attention maps of each word by the correspondence segmentation maps to control the location of objects. Additionally, DenseDiffusion [240] introduces a more extensive modulation method by devising multiple regularization, enhancing the precision and flexibility of layout control in score prediction. Furthermore, Chen et al. [244] introduce a soft refinement phase to dismiss the visual boundaries and enhance adjacent interactions.

For the attribute binding task, Ge et al. [249] study the image generation from enriched textual description and propose a region-based diffusion, which constrains the object-level description into the object area via an attention map. Similarly, Structure Diffusion [250] employs linguistic insights to manipulate the cross-attention map, aiming for more accurate attribute binding and improved image composition.

b) Feature injection in attention: By injecting additional key-value pairs into the attention module, the denoising process can dynamically incorporate visual information from the provided references. [241], [251], [252], [253], [253], [253], [254], [255], [256] Li et al. [252] inject object feature from one of the reference images into the inversion process of another image to realize object placement in self-attention. StyleAligned [253] is designed to produce a series of images that adhere to a given reference style. This method introduces a novel attention sharing mechanism within the self-attention layers, which facilitates the interaction between the features of individual images and those of an additional reference image. Such a design enables the generation process to consider and incorporate style elements from multiple images simultaneously. FreeControl [241] performs PCA to self-attention feature and replaces the principal components of feature in generation by reference components to control object appearance or spatial arrangement. Furthermore, feature injection can also be applied to cross-attention [257], [258]. Pick-and-draw [257] extracts cross-attention map in reference image inversion and generation process, and then injection the inversion feature into the generation process via Earth Movers Distance (EMD) algorithm.

c) Others: FreeU et al. [259] enhances diffusion image generation by strategically reweighting skip-connection and backbone feature contributions during inference—boosting coherence and fidelity without any additional training or parameters. Basu et al. [260] propose Mechanistic Localization in text-to-image models, demonstrating that knowledge of visual attributes (e.g., “style,” “objects,” “facts”) can be localized to a small subset of UNet layers, thereby enabling more efficient model editing.

To synthesize high-resolution images, MultiDiffusion [261] formulates an optimization problem that enforces each crop to remain consistent with its denoised counterpart. Although individual denoising steps may introduce conflicting directions,

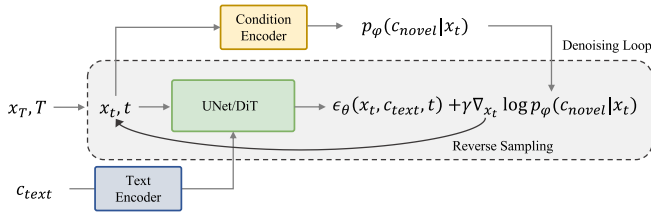


Fig. 6. Illustration of condition-guided conditional score estimation.

the method fuses them into a unified global denoising step, ultimately producing seamless and high-quality images. Zhou et al. [262] find that the padding is the pivotal mechanism to object arrangement, which is degraded in high-resolution generation. Based on that, they introduce a Progressive Boundary Complement, which creates dynamic virtual image boundaries inside the feature map to enhance position information propagation.

Meanwhile, from the noise initialization perspective, InitNo [263] first samples a lot of initial noise and then designs a cross-attention response score and the self-attention conflict score to evaluate them to find a better one.

B. Condition-Guided Score Estimation

While numerous methods adhere to the paradigm of conditional score prediction, some studies explore controllable generation by performing condition prediction from latents or intermediate features during the generative process, and then computing losses against the given conditions to provide gradient guidance for denoising [264], [264], [265], [266], [267], [268], [269], [270], as shown in Fig. 6.

LGP [265] stands as an early pioneer, which innovatively introduces a Latent Edge Predictor, designed to extrapolate sketch information from a series of intermediate features within a UNet architecture. It employs the degree of similarity between the condition sketch and predicted sketch to compute gradients, which are then utilized to guide the score estimation process. Its methodologies and insights have been a source of inspiration for numerous subsequent research endeavors in this field [243], [264], [268], [269]. Furthermore, Universal Guidance [266] and FreeDom [267] are proposed to leverage image-space off-the-shelf predictors to guide denoising. At each denoising step, it attains the clean image by one-step denoising to calculate the guidance gradient.

While the aforementioned methods require a condition predictor to backpropagate condition guidance, layout and segmentation guidance can also be directly estimated through attentionmap, eliminating the need for additional trained models [243], [257], [268], [269], [271], [272], [273], [274], [275]. For instance, BoxDiff [272] designs three spatial constraints (i.e., Inner-Box, Outer-Box, and Corner Constraints) to guide the denoising process. ZestGuide [269] leverages segmentation maps extracted from cross-attention layers, aligning generation with input masks through gradient-based guidance during denoising. To place the object at a specific position, VersaGen [273] calculates the loss from the object-token attention map and

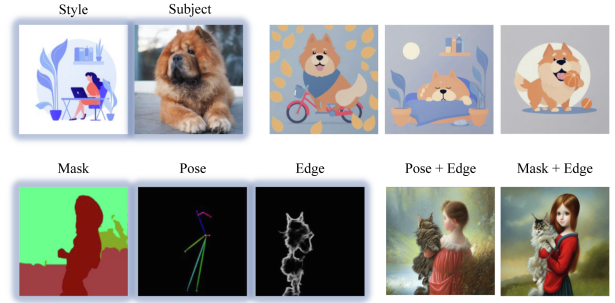


Fig. 7. Illustration of multi-conditioned generation. The condition is marked in blue background. Examples are sourced from [133], [280].

the given segmentation. Additionally, VODiff [276] studies the objects' visibility order and proposes visibility-order-aware loss.

Measuring the representation of textual concepts as a denoising guidance can help improve textual alignment [247], [277], [278]. Attend-and-Excite [277] (A&E) represents an early effort in this area, introducing an attention-based Generative Semantic Nursing (GSN) mechanism. This mechanism refines cross-attention units to more effectively ensure that all subjects described in the text prompt are accurately generated. EBAMA [279] extend A&E by introducing an attribute binding loss to address semantic misalignment. Additionally, SynGen [247] employs a unique methodology in text-to-image generation by first conducting a syntactic analysis of the text prompt. This analysis aims to identify entities and their modifiers within the prompt. Following this, SynGen utilizes a novel loss function designed to align the cross-attention maps with the linguistic bindings as indicated by the syntax.

IV. CONTROLLABLE GENERATION WITH MULTIPLE CONDITIONS

The multi-condition generation task aims to generate images under multiple conditions (see Fig. 7). In this section, we conduct a comprehensive overview of these methods from a technical perspective, categorizing them into joint training (Section IV-A), continual learning (Section IV-B), weight fusion (Section IV-C), attention-based integration (Section IV-D), and guidance composition (Section IV-E). Note that some of the other controllable generation methods also demonstrate multi-condition synthesis capability without dedicated designs [43], [64], [65], [135].

A. Joint Training

Designing a multi-condition framework and jointly training them is a simple yet effective route to realize multi-condition generation. These methods generally focus on multi-condition encoders and training strategies [105], [111], [111], [133], [185], [281], [282], [283], [284], [285].

Composer [281] projects all conditions (including text caption, depthmap, sketch, and etc.) into uniform-dimensional embeddings with the same spatial size as the noisy latent using stacked convolutional layers. It leverages a joint training strategy to generate images from a set of representations, where it uses

an independent dropout probability of 0.5 for each condition, a probability of 0.1 for dropping all conditions, and a probability of 0.1 for retaining all conditions. Additionally, Cocktail [133] proposes the controllable normalization method (ControlNorm), which has an additional layer to generate two sets of learnable parameters conditioned on all modalities. These two sets of parameters are used to fuse the external conditional signals and the original signals. UniCombine [282] designs a Conditional MMDiT Attention mechanism, where condition-specific LoRA modules mask the attention across different conditions to naturally support multiple conditions.

From a data perspective, SVDiff [111] utilizes a cut-mix-unmix mechanism for a multi-subject generation. It augments multi-concept data by a CutMix-like data augmentation and rewrites the correspondence text prompt. It also leverages an unmix regularization on cross-attention maps, ensuring text embeddings are only effective in the correspondence areas. This attention map constraint mechanism is also applied in FastComposer [185].

B. Continual Learning

Continual learning methods are generally proposed to address knowledge “catastrophic forgetting” in tuning-based conditional score prediction works [83], [286], [287], [288], [289]. Specifically, C-LoRA [83] is composed of a continually self-regularized LoRA in cross-attention layers. It utilizes the past LoRA weight deltas to regulate the new LoRA weight deltas by guiding which parameters are most available to be updated for continual concept learning. Moreover, L²DM [287] devises a task-aware memory enhancement module and an elastic-concept distillation module, which could respectively safeguard the knowledge of both prior concepts and each past personalized concept. It utilizes a rainbow-memory bank strategy to manage long-term and short-term memory and provide regularization samples to safeguard the knowledge in the personalization process. During training, the authors further propose a concept attention artist module and orthogonal attention artist module to update noised latent for better performance. STAMINA [288] introduces forgetting-regularization and sparsity-regularization in continual learning, avoiding forgetting learned concepts and ensuring no cost to storage or inference. ConceptGuard [289] combines shift embedding, concept-binding prompts, and memory preservation regularization to support new concepts.

C. Weight Fusion

In the realm of adapting T2I diffusion models to novel conditions via fine-tuning, weight fusion presents itself as an intuitive approach for merging multiple conditions. These methods focus on achieving a cohesive blend of weights that incorporates each condition while ensuring that the controllability of individual conditions is retained. The goal is to seamlessly integrate various conditional aspects into a unified model, thereby enhancing its versatility and applicability across diverse scenarios. This requires a delicate balance between maintaining the integrity of each condition’s influence and achieving an effective overall synthesis.

Since personalized conditions usually represent UNet’s weight or text embeddings, weight fusion is an intuitive and effective way to generate images under multiple personalized conditions. Specifically, Cones [290] further fine-tunes the concept neurons after personalization for better generation quality and multi-subject generating capability. Custom Diffusion [71] introduces a constrained optimization method to merge fine-tuned key and value matrices, as follows:

$$\begin{aligned} \hat{W} &= \arg \min_W \|WC_{\text{reg}} - W_0C_{\text{reg}}\|_F \\ \text{s.t. } WC^T &= V, \text{ where } C = [c_1 \dots c_N]^T \\ \text{and } V &= [W_1c_1^T \dots W_Nc_N^T]^T \end{aligned} \quad (18)$$

where $\{W_{n,l}^k, W_{n,l}^v\}_{n=1}^N$ represent the corresponding updated key and value matrices for added N concepts and C_{reg} is a randomly sampled text features for regularization. The objective of (18) is intuitively designed to ensure that the words in the target captions are consistently aligned with the values derived from the concept matrices that have undergone fine-tuning. Similarly, Mix-of-Show [82] introduces the gradient fusion, updating weight W by $W = \arg \min_W \sum_{i=1}^n \|(W_0 + \Delta W_i)X_i - WX_i\|_F^2$ where X_i represents the input activation of the i -th concept, and $\|\cdot\|_F$ denotes the Frobenius norm. To integrate subject-centric and style-centric conditions, ZipLoRA [280] merges LoRA-style weights by minimizing the difference between subject/style images generated by the mixed and original LoRA models and the cosine similarity between the columns of content and style LoRAs. Po et al. [291] present orthogonal adaption to replace LoRA in fine-tuning, encouraging the customized models to have orthogonal residual weights for efficient fusion.

D. Attention-Based Integration

Attention-based integration methods modulate attention maps to strategically position subjects within the synthesized image, allowing for precise control over where and how each condition is represented in the final composition [82], [292], [293].

For example, Cones2 [292] edits cross-attention map by $\text{EditedCA} \leftarrow \text{Softmax}(CA \oplus \{\eta(t) \cdot M_{s_i} | i = 1, \dots, N\})$, where $\oplus$ denotes the operation that adds the corresponding dimension of cross-attention map CA and pre-defined layout M and $\eta(t)$ is a concave function controlling the edit intensity at different timestep t . Similarly, Mix-of-Show [82] employs a regionally controllable sampling method, integrating global prompt and multiple regional prompts with pre-defined masks in cross-attention.

E. Guidance Composition

Guidance composition is an integration mechanism for synthesizing images under multiple conditions, integrating the independent denoising results of each condition [42], [294], [295], [296], [297], [298]. This process is mathematically represented

as:

$$\hat{\epsilon}(z_t, c_1, \dots, c_N) = \sum_{i=1}^K w_i \cdot \mathcal{M}_i \cdot \epsilon(z_t, c_i) \quad (19)$$

where $\epsilon(z_t, c_i)$ denotes the guidance of each condition, while w_i and $\mathcal{M}_i$ are the respective weights and spatial mask used to integrate these results.

To integrate multiple concepts, Decompose and Realign [296] obtains the corresponding $\mathcal{M}_i$ by their cross-attention map. Similarly, Face-diffuser [295] presents a saliency-adaptive noise fusion method to combine results from a text-driven diffusion model and a proposed subject-augmented diffusion model.

Besides, to realize controllable generation in user-specific domain, Cao et al. [42] train a null-text UNet to provide domain guidance and utilize the original diffusion prior to provide control guidance.

V. UNIVERSAL CONTROLLABLE TEXT-TO-IMAGE GENERATION

Beyond approaches tailored to specific types of conditions, there exist universal methods designed to accommodate arbitrary conditions in image generation. These methods are broadly categorized into two groups based on their theoretical foundations: universal conditional score prediction framework and universal condition-guided score estimation.

A. Universal Conditional Score Prediction Framework

Universal conditional score prediction framework involves creating a framework capable of encoding any given conditions and utilizing them to predict the noise at each timestep during the image synthesis process. This approach provides a universal solution that adapts flexibly to diverse conditions. By integrating the conditional information directly into the generative model, this method allows for the dynamic adaptation of the image generation process in response to a wide array of conditions, making it versatile and applicable to various image synthesis scenarios.

DiffBlender [299] is proposed to incorporate conditions from diverse types of modalities. It categorizes conditions into multiple types to employ different techniques for guiding generation. First, image-form conditions, which contain spatially rich information, are injected in ResNet Blocks [300]. Then, spatial conditions, including grounding box and keypoints, are passed through a local self-attention module to accurately locate the desired positions of synthesized results. Moreover, non-spatial conditions like color palette and style are concatenated with textual tokens through a global self-attention module and then fed into cross-attention layers. Additionally, Emu2 [301] leverages a large generative multimodal model with 37 billion parameters for task-agnostic in-context learning to construct a universal controllable T2I generation framework.

B. Universal Condition-Guided Score Estimation

Other approaches utilize condition-guided score estimation to incorporate various conditions into the text-to-image diffusion

models. The primary challenge lies in obtaining condition-specific guidance from the latent during the denoising process.

Universal Guidance [266] observes that the reconstructed clean image proposed in the denoising diffusion implicit model (DDIM) [302] is appropriate for a generic guidance function to provide informative feedback to guide the image generation. Given any condition c and off-the-shelf predictor f , the denoising process is guided by:

$$\hat{\epsilon}_\theta(z_t, t) = \epsilon_\theta(z_t, t) + s(t) \cdot \nabla_{z_t} \mathcal{L}(c, f(\hat{z}_0)) \quad (20)$$

where $\hat{z}_0$ is the predicted clean image following [302]:

$$\hat{z}_0 = \frac{z_t - (\sqrt{1 - \alpha_t})\epsilon_\theta(z_t, t)}{\sqrt{\alpha_t}} \quad (21)$$

UG employs various predictors, including CLIP [34] (for text or style conditions), segmentation network [303] (for segmentation map conditions), face recognition model [304], [305] (for identity conditions), and object detector [306] (for bounding box conditions), in experiments to exhibit conditional generation capabilities with various conditions.

Similar to Universal Guidance [266], FreeDom [267] leverages off-the-shelf predictors to construct time-independent energy functions to guide the generation process. It also develops the efficient time-travel strategy, taking the current intermediate result z_t back by j steps to z_{t+j} and resampling it to the t -th timestep. This mechanism solves the problem of misalignment with conditions on large data domains, e.g. ImageNet [307].

While above mentioned condition-guided sampling approaches leverage off-the-shelf models and one-step estimation procedure to predict condition-related conditions, Pan et al. [308] present Symplectic Adjoint Guidance (SAG) in two inner stages, where SAG first estimate the clean image via n function calls and then uses the symplectic adjoint method to obtain the gradients accurately.

VI. APPLICATIONS

In this section, we focus on innovative methods that utilize novel conditions in the generation process to address specific tasks. By emphasizing these pioneering approaches, we aim to highlight how conditional generation is not only reshaping the landscape of content creation but also broadening the horizons of creativity and functionality in various fields. The subsequent discussions will provide insights into the transformative impact of these models and their potential in diverse applications. We illustrate the example of the applications in Fig. 8.

A. Image Manipulation

Advancements in the control of pre-trained text-to-image diffusion models have allowed for more versatile image editing techniques. For instance, inspired by DreamBooth [65], SINE [316] constructs the text prompt for fine-tuning the pretrained text-to-image model by the source image as “a photo/painting of a [*] [class]” and edits the image by a novel adapter-based classifier-free guidance. Moreover, the versatility of control conditions further enhances the editing process by integrating conditions beyond mere text. For example, Choi et al.

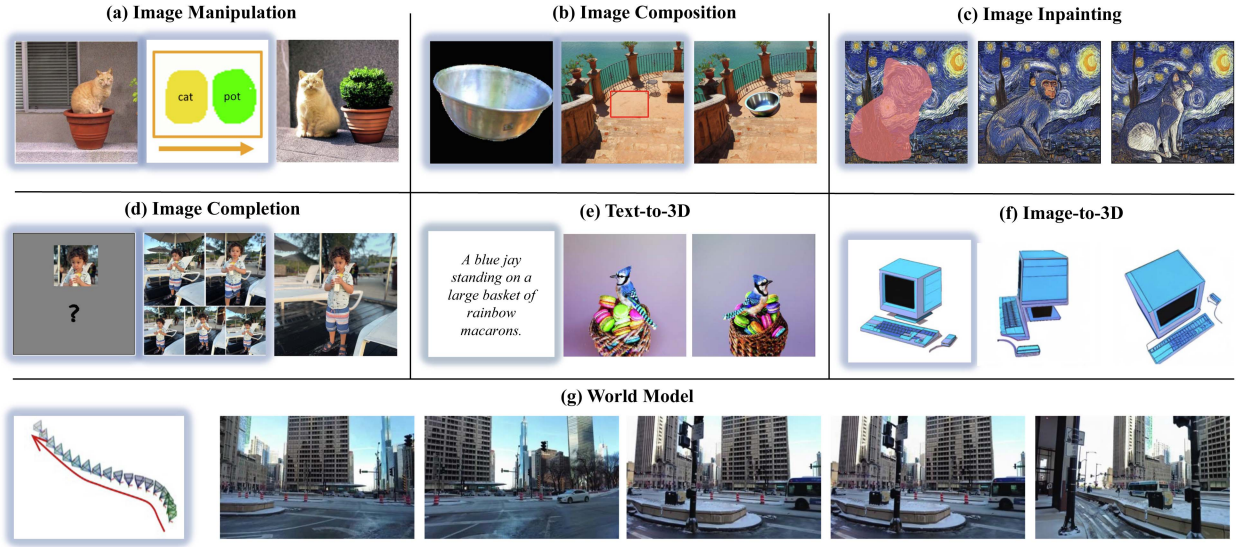


Fig. 8. Illustration of the application of controllable text-to-image generation. The condition is marked in blue background. Examples are sourced from [309], [310], [311], [312], [313], [314], [315].

[317] customize the diffusion model to employ specific elements from the reference image as editing criteria, such as substituting the cat in the source image with the cat's appearance in the reference image. Recently, Zhou et al. [318] modify the score estimation in multi-turn editing, introducing a dual-objective Linear Quadratic Regulators (LQR) to effectively mitigate error accumulation.

B. Image Completion and Inpainting

The advancement of flexible control mechanisms has also significantly expanded the capabilities in the field of image inpainting and completion. Specifically, DreamInpainter [311] utilizes a subject-driven generation approach to personalize the filling of masked areas with the aid of reference images. Besides, Realfill [312] takes similar methods that employ reference images to facilitate realistic and coherent image completions. Moreover, by multiple condition controlling, Uni-inpaint [319] integrates a diverse set of control conditions such as text descriptions, strokes, and exemplar images to simultaneously direct the generation within the masked regions.

C. Image Composition

Image composition is a challenging task that involves multiple complex image process stages like color harmonization, geometry correction, shadow generation, and so on. While the strong prior in large-scale pre-trained diffusion model can address the problem in a unified manner. Through adding adapters to control the pre-trained text-to-image diffusion model, Object-Stitch [310] presents an object composition framework that can handle multiple aspects such as viewpoint, geometry, lighting, and shadow. Moreover, DreamCom [320] customizes the text-to-image model on several foreground object images to enhance the object details' preservability. Besides, by inserting the task indicator vector into U-Net to control the generating process,

ControlCom [321] proposes a controllable image composition method that unifies four composition-related tasks with an indicator vector.

D. Text/Image-to-3D Generation

Text/image-to-3D task aims to reconstruct 3D representations from text descriptions or images (pairs). Recent advancements in text/image-to-3D generation represent a significant milestone with the development of Score Distillation Sampling (SDS) loss. This innovative approach, introduced by DreamFusion [322], marks a successful adaptation of large-scale 2D diffusion models for 3D generation. Through SDS, the control method of the text-to-image model can be transferred to text-to-3D generation. Typically, DreamBooth3D [323] combines DreamBooth [65] and DreamFusion [322] that personalizes text-to-3d generative models from a few captured images of a subject. Similarly, some approaches [324], [325], [325], [326] adapt ControlNet [43] to the SDS process, enabling the control of 3D generation through spatial signals (e.g., depth map, sketch).

E. World Model

The condition-injection mechanisms provide effective support for the development of video-generation-based world models. A representative example is camera-controlled video generation [315], [327], [328], [329], [330], [331], [332], which focuses on aligning sequences of camera parameters with the diffusion-based video generation process. For instance, Re-CamMaster [332] incorporates camera parameters into the original DiT blocks via frame-dimension conditioning. Similarly, AC3D [330] introduces camera information through lightweight 128-dimensional DiT-SX blocks. In addition, several approaches tackle this problem in a training-free manner [333], [334], [335]. Typically, WorldForge [335], which leverages 3D/4D foundation models [336] to project video

frames into a static point cloud. The point cloud is then adjusted according to camera trajectories and subsequently used as guidance for video generation.

VII. CONCLUSION

In this comprehensive survey, we delve into the realm of conditional generation with text-to-image diffusion models, unveiling the novel conditions incorporated in the text-guided generation process. Initially, we equip readers with foundational knowledge, introducing the denoising diffusion probability models, prominent text-to-image diffusion models, and a well-structured taxonomy. Subsequently, we reveal the mechanisms of introducing novel conditions into T2I diffusion models. Then, we present a summary of previous conditional generation methods and analyze them in terms of theoretical foundations, technical advancements, and solution strategies. Furthermore, we explore the practical applications of controllable generation, underscoring its vital role and immense potential in the era of AI-generated content. This survey aims to provide a comprehensive understanding of the current landscape of controllable T2I generation, thereby contributing to the ongoing evolution and expansion of this dynamic research area.

Although controllable generation with text-to-image diffusion models has achieved remarkable progress, several promising directions remain open for future exploration:

(1) *Towards a universal and cross-modal control paradigm:* Existing controllable diffusion models are often tailored to specific conditions or tasks. Future research could focus on developing unified and generalizable control frameworks capable of flexibly accommodating diverse forms of conditions, including spatial, semantic, and multimodal inputs, within a single generative system. Extending controllability beyond text-to-image synthesis to other modalities such as audio, video, and 3D generation will further enhance the model's adaptability and cross-modal reasoning ability, paving the way toward general multimodal intelligence.

(2) *Building world models with controlling mechanisms:* The condition-injection and controllability principles of diffusion models provide a strong foundation for constructing world models based on video generation. Future studies could explore how diffusion-based systems simulate dynamic, camera-controllable environments while maintaining temporal-spatial consistency. Such world models will play a crucial role in connecting generative modeling, embodied AI, and interactive virtual environments.

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